

# Fate of heavy metals in a strongly acidic shooting-range soil: small-scale metal distribution and its relation to preferential water flow

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## Summary – Zusammenfassung

To assess the mobility of Pb and associated metals in a highly contaminated shooting range soil (Losone, Ticino, Switzerland), we investigated the spatial distribution of the metals and their relation to preferential water flow paths. A 2.2 m<sup>2</sup> plot located 40 m behind the stop butt was irrigated with a solution containing bromide and Brilliant Blue, a slightly sorbing dye. A soil profile 50 cm in width was sampled down to 80 cm with a spatial resolution of 2.5 cm, resulting in 626 samples. Concentrations of elements ( $12 \leq Z \leq 92$ ) were determined by energy-dispersive X-ray fluorescence spectrometry, and Brilliant Blue concentrations were determined with a chromameter. In the acidic (pH 3), organic matter-rich, well drained Dystric Cambisol, maximum concentrations of 80.9 g kg<sup>-1</sup> Pb, 4.0 g kg<sup>-1</sup> Sb, and 0.55 g kg<sup>-1</sup> Cu were measured in the topsoil. Within 40 cm soil depth, however, Pb, Sb, and Cu approached background concentrations of 23 mg kg<sup>-1</sup>, 0.4 mg kg<sup>-1</sup>, and 9.4 mg kg<sup>-1</sup>, respectively. The even horizontal distribution and the steep gradient along soil depth indicate tight metal binding in the topsoil, and a fairly homogeneous transport front. In contrast, water flow through the profile was highly heterogeneous. In the uppermost 20 cm, preferential flow was initiated by heterogeneous infiltration at the soil surface, but had no influence on metal distribution. Below 20 cm, however, preferential flow originated from larger tree roots, and metal concentrations were significantly elevated along these macropores. Spatial distributions of Pb, Sb, and Cu were similar, suggesting that all three metals are strongly retained in the topsoil and transported along preferential water flow paths in the subsoil.

## Verhalten von Schwermetallen in einem stark sauren Schießplatzboden: Kleinräumige Metallverteilung in Beziehung zu präferenziellem Wasserfluss

Um die Mobilität von Pb und assoziierten Metallen in einem hoch belasteten Schießplatzboden (Losone, Tessin, Schweiz) zu erfassen, untersuchten wir die räumliche Verteilung der Metalle und ihre Beziehung zu präferenziellen Fließwegen. Eine 2.2 m<sup>2</sup> große Parzelle 40 m hinter dem Kugelfang wurde mit einer Lösung aus Bromid und Brilliant Blue, einem schwach sorbierenden Farbstoff, beregnet. Ein 50 cm breites und 80 cm tiefes Bodenprofil wurde mit einer räumlichen Auflösung von 2.5 cm beprobt, wobei insgesamt 626 Proben anfielen. Die Konzentrationen der Elemente  $12 \leq Z \leq 92$  wurde mittels energie-dispersiver Röntgenfluoreszenz-Spektrometrie bestimmt, die Konzentration an Brilliant Blue mit einem Farbmessgerät. Die saure (pH 3), humusreiche, gut durchlässige Locker-Braunerde (Dystric Cambisol) wies im Oberboden maximale Konzentrationen von 80.9 g kg<sup>-1</sup> Pb, 4.0 g kg<sup>-1</sup> Sb und 0.55 g kg<sup>-1</sup> Cu auf. Innerhalb von 40 cm Bodentiefe sanken die Konzentrationen jedoch auf Hintergrundwerte von 23 mg kg<sup>-1</sup> Pb, 0.4 mg kg<sup>-1</sup> Sb und 9.4 mg kg<sup>-1</sup> Cu ab. Die gleichmäßige horizontale Verteilung und der steile Gradient entlang der Bodentiefe weisen auf eine feste Metallbindung im Oberboden und eine relativ homogene Transportfront hin. Im Gegensatz dazu war der Wasserfluss jedoch außerordentlich heterogen. In den obersten 20 cm fand präferenzieller Fluss durch heterogene Infiltration an der Bodenoberfläche statt. Dieser hatte jedoch keinen Einfluss auf die Schwermetallverteilung. Unterhalb von 20 cm wurde präferenzieller Fluss durch größere Baumwurzeln verursacht. In diesen Makroporen waren die Metallkonzentrationen signifikant gegenüber der Bodenmatrix erhöht. Aus der deckungsgleichen räumlichen Verteilung von Pb, Sb und Cu kann geschlossen werden, dass alle drei Metalle im Oberboden stark zurückgehalten und im Unterboden entlang von präferenziellen Fließwegen transportiert werden.

**Keywords:** preferential flow / Pb / Sb / Cu / soil heterogeneity / shooting range

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## 1 Introduction

Lead contamination of soils may stem from various sources, including mining and smelting of Pb ores, deposition or recycling of spent Pb batteries, waste incineration, and combustion of leaded gasoline (Davies, 1995; Manceau et al., 1996). Deposition of Pb bullets and pellets in military and recreational outdoor shooting ranges is another important source, leading to Pb concentrations of up to 50,000 mg kg<sup>-1</sup> in the topsoil around targets and stop butts (Fahrenhorst and Renger, 1990; Jørgensen and Willems, 1987; Lin, 1996; Manninen and Tanskanen, 1993). In Switzerland there are more than 2000 rifle-shooting ranges, and 400 to 500 tons of Pb are annually deposited (Anonymous, 1997).

Metallic Pb has a relatively low solubility, and the small amount of dissolved Pb<sup>2+</sup> is usually tightly bound by roots, soil organic matter, and Mn and Fe oxides (Adriano, 1986; Manceau et al., 1996). Consequently, the retention time of Pb in contaminated soils is long, with half-life estimates ranging from 740 to 5900 years (Kabata-Pendias and Pendias, 1992). Thus, in spite of high concentrations and a substantial toxicity of Pb, the risk of these sites has been considered relatively low. In fact, the highest risk emanates from digestion of bullets or contaminated soil by playing children (Wixson and Davies, 1994), grazing cattle (Braun et al., 1997), and wildlife (Lewis et al., 2001), which may be easily prevented. Many investigations confirmed the low mobility of Pb in shooting range soils. Even at the most heavily contaminated areas around targets and stop butts, concentrations of total Pb rapidly declined within a few decimeters distance from the surface (Astrup et al., 1999; Chen et al., 2002; Lin et al., 1995; Turpeinen et al., 2000). Lead dissolved in soil water as well as water-extractable Pb were low (Fahrenhorst and Renger, 1990; Turpeinen et al., 2000).

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However, a few investigations indicate a potential risk of Pb migrating towards groundwater resources. Murray et al. (1997) and Desaulles and Dahinden (1995) found elevated Pb concentrations in soil depths of one meter. In a leaching study with undisturbed soil columns from a shooting range, Rooney and McLaren (1999) measured Pb concentrations in the effluent of up to 3400  $\mu\text{g l}^{-1}$ . Furthermore, a common additive to Pb bullets, Sb, has been found at concentrations of up to 155  $\mu\text{g l}^{-1}$  in soil solution, posing a risk of its own (Fahrenhorst and Renger, 1990). The processes leading to a potential mobilization of Pb and other heavy metals are still not well understood and are the subject of current research.

Once a bullet has penetrated into the soil, the surface of the metallic Pb core is rapidly oxidized forming a crust of secondary Pb phases. For a wide range of soil conditions including pH values from 3 to 7.5, the most prevalent mineral in the crust is hydrocerussite ( $\text{Pb}_3(\text{CO}_3)_2(\text{OH})_2$ ), besides minor amounts of cerussite ( $\text{PbCO}_3$ ), anglesite ( $\text{PbSO}_4$ ), pyromorphite ( $\text{Pb}_5(\text{PO}_4)_3\text{Cl}$ ), massicot ( $\beta\text{-PbO}$ ), and plattnerite ( $\alpha\text{-PbO}_2$ ) (Chen et al., 2002; Fahrenhorst and Renger, 1990; Jørgensen and Willems, 1987; Lin, 1996). Compared to the other minerals, the solubility of hydrocerussite is high, and increases with decreasing pH ( $\text{Pb}_3(\text{CO}_3)_2(\text{OH})_2 + 2\text{H}^+ \leftrightarrow 3\text{Pb}^{2+} + 2\text{H}_2\text{O} + 2\text{CO}_3^{2-}$ ;  $\log K = -18.8$ ). Correspondingly, surface water which had been collected in small ditches containing hydrocerussite-coated bullets reached Pb concentrations of 470  $\mu\text{g l}^{-1}$  at a pH of 5.5 (Craig et al., 1999). In soil, however, the Pb concentration is drastically reduced by several processes.  $\text{Pb}^{2+}$  forms tight binding complexes with organic matter (Davies, 1995; Davies et al., 1997; Manceau et al., 1996), which has been confirmed for shooting-range soils by selective extraction methods (Basunia and Landsberger, 2001; Bruell et al., 1999; Jørgensen and Willems, 1987; Manninen and Tanskanen, 1993). Fe and Mn hydroxides sorb  $\text{Pb}^{2+}$  via formation of inner-sphere sorption complexes (Balikungeri and Haerdi, 1988; Davies, 1995; Manceau et al., 1992; McKenzie, 1980; Scheinost et al., 2001). Especially in farm soils with elevated P concentrations, Pb may precipitate as pyromorphite, which has an extremely low solubility at common soil pH (Cotter-Howells et al., 1994; Lindsay, 1979). Finally, roots are very strong sinks for  $\text{Pb}^{2+}$ , either by sorption to cell walls or by precipitation of insoluble Pb salts (Adriano, 1986; Sarret et al., 1998), explaining the reduction of Pb in soil water by pine seedlings in comparison to a plant-free soil (Turpeinen et al., 2000).

Therefore, as long as rain water filtrates homogeneously through the soil matrix, migration of soluble Pb towards the groundwater seems unlikely. Thus, the few observed incidents of elevated Pb concentration in soil effluent and in greater soil depths indicate a different transport mechanism. One possibility is transport of Pb by preferential water flow. Preferential flow significantly enhanced the transport of reactive contaminants like pesticides (Flury et al., 1995) and led to an enrichment of radionuclides in the preferential flow paths (Bundt et al., 2000). Since preferential flow seems to be common in many different soils (Flury et al., 1994; Miyazaki, 1993), it may lead to a local enrichment of Pb in the subsoil, which is difficult to detect by conventional sampling techniques. In fact, Murray et al. (1997) suggested that high

Pb concentrations in the subsoil could be related to cracks in the soil profile which may act as travel paths for preferential water flow and Pb.

The objective of our work was to investigate the relation between preferential water flow and the distribution of Pb and associated metals in a shooting-range soil profile. Bundt et al. (2000; 2001a; 2001b) investigated correlations between preferential water flow and a range of soil properties by irrigating a soil plot with a dye tracer, and compared the soil properties of stained versus unstained regions. While this method allows testing of the *a priori* hypothesis that preferential water flow does or does not influence the respective soil properties, it is not possible to determine whether preferential water flow is the only factor influencing the Pb distribution in a soil profile or not. Therefore, we used an alternative approach by sampling a regular grid with a spacing of 2.5 cm across a soil profile after irrigating the soil with two tracers of different sorptivity, Brilliant Blue and bromide. Two-dimensional maps of element and tracer distributions then allowed (1) to investigate the spatial heterogeneity of Pb and other contaminants like Sb, Cu, and Zn at the centimeter scale and (2) to compare the heavy metal distribution patterns with that of preferential water flow.

## 2 Material and methods

### 2.1 Study site

The investigation was carried out on the Swiss Army 300-m rifle shooting range of Losone, a village close to the northern end of Lake Maggiore in southern Switzerland (8°45'44"E/46°10'21"N). An undisturbed soil, located about 50 m behind the target area and 40 m behind the stop butt, was selected for the investigation. The site is situated 250 m above sea level on a north-east facing slope, which is covered by chestnut (*Castanea sativa*) low forest with some occasional birch (*Betula pendula*) and oak trees (*Quercus pubescens*) ("palina" type locality; Ellenberg, 1986). The mean annual temperature is 15 °C and the mean annual precipitation is 1840 mm. The 300-m shooting range was built in 1960 and has been used intensively until 1998 for both, military and civil purposes. During this time, the annual input of bullets decreased from approximately 800 kg to 500 kg. Since 1998, the shooting range has been used for recreational purposes only, which reduced the deposition of bullets to 40 kg  $\text{a}^{-1}$ . Two types of bullets have been used predominantly (GP 11 and GP 90), both with a similar composition of approximately 0.75 g  $\text{g}^{-1}$  Pb, 0.20 g  $\text{g}^{-1}$  Fe, 25 g  $\text{kg}^{-1}$  Cu, 15 g  $\text{kg}^{-1}$  Sb, 5 g  $\text{kg}^{-1}$  Ni, and < 0.1 g  $\text{kg}^{-1}$  As (Anonymous, 1997).

Figure 1a shows a photograph of the soil profile, a well drained Dystric Cambisol (Lockerbraunerde) characterized by low pH, low bulk density, and a 50 cm thick topsoil (Tab. 1). The organic matter content decreases continuously with depth, but even at 50 cm depth the soil is dark brown with a Munsell color of 10 YR 4/2. The accumulation of organic matter has been explained by the tannin-rich chestnut litter and the high content of free Al of the soil, both components forming stable organo-metallic complexes (Blaser et al.,

**Table 1:** Chemical and physical soil properties.**Tabelle 1:** Chemische und physikalische Bodeneigenschaften.

| Soil horizon          | Depth (cm) | Bulk density (Mg m <sup>-3</sup> ) | pH (0.01 M CaCl <sub>2</sub> ) | ECEC (mmol <sub>c</sub> kg <sup>-1</sup> ) | Base saturation (% ECEC) | C/N ratio | C <sub>org</sub> (g kg <sup>-1</sup> ) | Mn (g kg <sup>-1</sup> ) | Fe (g kg <sup>-1</sup> ) | Ni (g kg <sup>-1</sup> ) | Cu (mg kg <sup>-1</sup> ) | Zn (mg kg <sup>-1</sup> ) | Sb (mg kg <sup>-1</sup> ) | Pb (mg kg <sup>-1</sup> ) |
|-----------------------|------------|------------------------------------|--------------------------------|--------------------------------------------|--------------------------|-----------|----------------------------------------|--------------------------|--------------------------|--------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| <b>L</b>              | –5 to 0    | n.d.                               | 3.2                            | n.d.                                       | n.d.                     | 17.7      | 331                                    | 0.18                     | 16                       | 43.9                     | 148.9                     | 87.0                      | 676                       | 12533                     |
| <b>Ah<sub>1</sub></b> | 0 to 10    | 0.47                               | 3.6                            | 119                                        | 12                       | 16.4      | 148                                    | 0.29                     | 30                       | 38.9                     | 95.8                      | 65.1                      | 30                        | 2908                      |
| <b>Ah<sub>2</sub></b> | 10 to 50   | 0.59                               | 4.2                            | 62                                         | 5                        | 18.1      | 89                                     | 0.38                     | 31                       | 35.3                     | 12.1                      | 57.4                      | 1.9                       | 85                        |
| <b>Bv</b>             | 50 to 60   | 0.95                               | 4.7                            | 33                                         | 63                       | 17.9      | 27                                     | 0.52                     | 33                       | 51.0                     | 9.8                       | 66.5                      | < 0.4                     | 24                        |
| <b>Cv</b>             | 60 to 70   | 1.57                               | 4.8                            | 36                                         | 84                       | 16.8      | 13                                     | 0.55                     | 31                       | 64.0                     | 13.4                      | 58.5                      | < 0.3                     | 22                        |

n.d.: not determined

1997). In fact, exchangeable Al accounts for up to 94 % of the effective cation exchange capacity (ECEC, Tab. 1). The low ECEC is due to the lack of clay (less than 1 % in the top 50 cm) and the low pH. The main rooting zone is largely confined to the topsoil horizons (Ah<sub>1</sub> and Ah<sub>2</sub>) with only a few roots in the Bv horizon. Neither earthworms nor burrows were observed. The light brown Bv horizon (10 YR 5/3) has a sandy-loamy texture and a variable thickness. The Cv horizon (1 Y 6/2) consists of gravelly moraine sediment. Rocks are soft and crumbly and coated with a red layer of iron oxides indicative of intense weathering. The mineralogical composition, as determined by X-ray diffraction, is quartz, plagioclase, vermiculite, microcline, hornblende, chlorite, muscovite, and brookite.

## 2.2 Tracer irrigation

To identify the preferential water flow paths, we selected two tracers with different reactivities, bromide and Brilliant Blue. Bromide, a non-sorbing, conservative tracer was applied as CaBr<sub>2</sub> · 2 H<sub>2</sub>O at a concentration of 12 g l<sup>-1</sup> (50 mmol l<sup>-1</sup>). This concentration, which is about one order of magnitude higher than commonly employed (Flury et al., 1995), had to be used to get significant contrast between Br from the tracer and Br from the natural, pedogenic background (20 to 50 mg kg<sup>-1</sup>). Brilliant Blue FCF, a dye tracer, is often used to stain preferential flow paths because of its mobility, visibility, and low toxicity (Flury et al., 1994; Forrer et al., 2000). In comparison to bromide it is more retarded by soil, with distribution coefficients varying between 0.2 and 5.8 l kg<sup>-1</sup> for different soils (Flury and Flühler, 1995). Brilliant Blue was applied at a concentration of 4 g l<sup>-1</sup> (5 mmol l<sup>-1</sup>). Both tracers were dissolved in 100 l of local drinking water.

The tracer solution was evenly sprinkled with a watering can onto a 2.2 m<sup>2</sup> plot (45 l m<sup>-2</sup>) without ponding the surface. After 20 h, a pit was excavated and a vertical soil profile of ca. 1 m<sup>2</sup> in the center of the irrigated area was prepared.

## 2.3 Soil sampling

The sampling at the centimeter scale was performed with small plastic boxes (2.2 x 2.2 x 2.2 cm<sup>3</sup>) with a net volume of

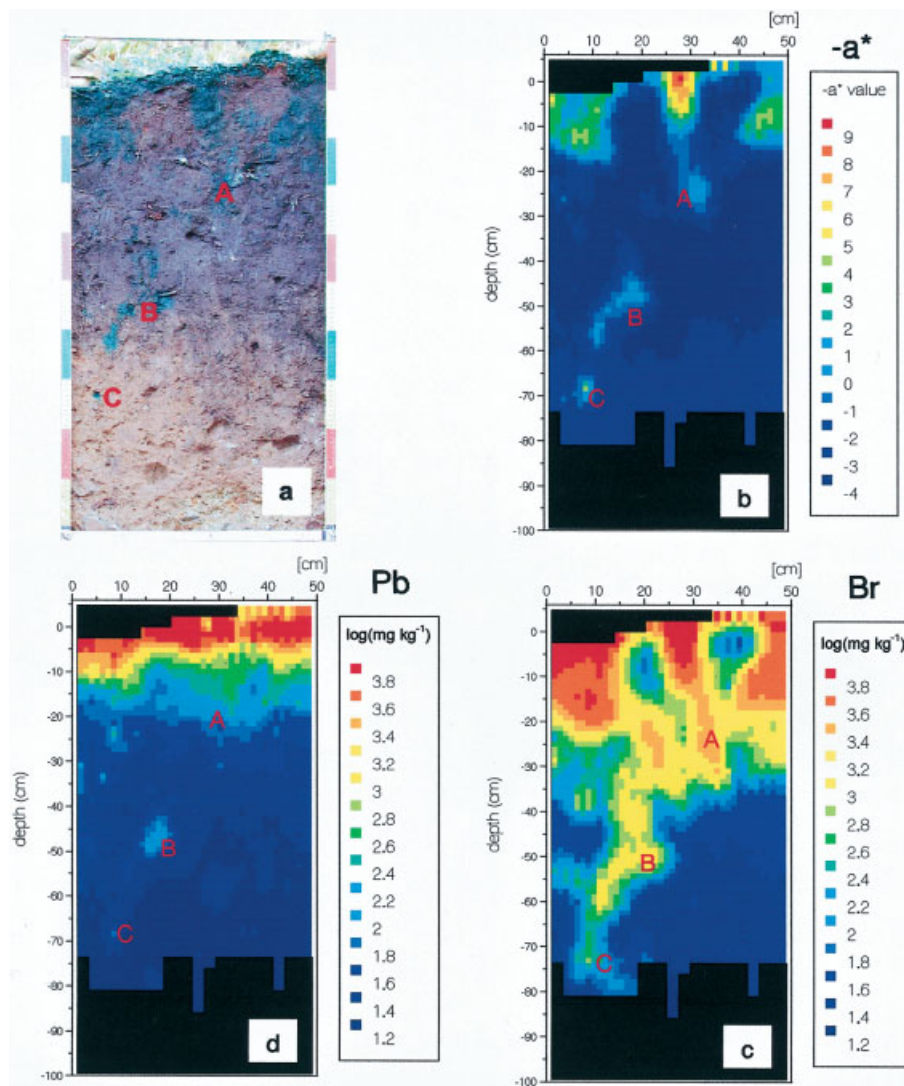
8 cm<sup>3</sup>. The boxes were pushed into the profile wall and lined up as closely to each other as possible, achieving an average grid size of 2.5 x 2.5 cm<sup>2</sup> across the profile, and a sampling depth of 2.2 cm perpendicular to the profile plane. A total of 626 boxes was sampled, covering an area of approximately 0.5 m in horizontal and 0.8 m in vertical direction. In addition, bulk samples were taken every 10 cm for standard soil analyses, and undisturbed soil cores (1 l volume) were collected at 4 depths in 2 replicates to measure bulk density (Tab. 1).

## 2.4 Chemical and physical soil analysis

The elemental composition of the small box samples and the bulk samples was determined by energy-dispersive X-ray fluorescence spectrometry. The fine earth fraction of the samples was air-dried and ground to a particle size of less than 60 µm, using a vibratory disk mill (Retsch). 2 to 4 g of soil were thoroughly mixed with a binding component (C-Wachs, Hoechst) at a fixed ratio (soil/wax = 40/9), pressed into pellets (4 cm diameter), and analyzed using an energy dispersive X-ray spectrometer (Spectro X-LAB 2000). By using polarized radiation, a series of 5 secondary targets, and an *ab initio* method to improve deconvolution of the spectra, a lower detection limit of 0.5 mg kg<sup>-1</sup> could be routinely achieved for Pb, Sb, Zn, and Ni. Accuracy was tested with certified standards.

The concentration of Brilliant Blue was determined semi-quantitatively by measuring the color of the ground soil samples with a Chroma Meter (CR-300, Minolta). Colors were expressed in the CIE Lab color system with Cartesian notation (Scheinost and Schwertmann, 1999). The coordinate -a\*, representing the greenness, was used to discriminate the color of the dye from the color of other soil components.

Soil pH was measured potentiometrically in 0.01 M CaCl<sub>2</sub>. Exchangeable cations were extracted with 1M NH<sub>4</sub>NO<sub>3</sub> for 24 h on a shaker (soil/solution=1/25) and measured by flame atomic absorption spectrometry (Varian SpectrAA 220FS). Carbon and N contents were measured with a CHNS analyzer (Leco CHNS 932).



**Figure 1:** (a) Photograph of the sampled soil profile with water infiltration and transport paths stained with Brilliant Blue, (b) 2-dimensional distribution of Brilliant Blue (expressed as greenness color coordinate,  $-a^*$ ), (c) Br, and (d) Pb. Spots "A", "B", and "C" are referred to in the text.

**Abbildung 1:** (a) Photo des Bodenprofils mit blau eingefärbten Wasserinfiltrations- und Transportpfaden, (b) zwei-dimensionale Verteilung von Brilliant Blue (gemessen als Grün-Komponente  $-a^*$ ), (c) Br und (d) Pb. Auf die mit "A", "B" und "C" gekennzeichneten Punkte wird im Text Bezug genommen.

## 2.5 Statistical Analysis and Data Mapping

Statistical analyses were performed with the software Statistica (StatSoft Inc. Tulsa, version 5.1). Data were tested for normality and log-transformed when necessary. Data maps were produced with CoPlot 3.084 (CoHort Software). A grid transformation was performed to account for missing data caused by loss of samples due to rocks and large roots. Simultaneously, the number of grids was doubled to enhance the visual resolution to 1.25 cm. Data points were interpolated by weighting the eight nearest neighbors by their inverse distances.

## 3 Results and discussion

### 3.1 Water infiltration and preferential water flow

Fig. 1 shows a photograph of the soil profile in comparison with a map of Brilliant Blue derived from the box sampling. In the photograph, blue-green spots due to accumulation of Brilliant Blue can be discerned even in the dark topsoil

horizon. The uppermost 5 cm of the soil profile are uniformly stained, but below this litter horizon the infiltrating solution branched into three distinct pathways. This heterogeneous infiltration can be explained by inhomogeneous distribution of the hydrophobic litter layer and by small depressions at the soil surface, which collected and infiltrated the predominant part of the irrigated water, while intermediate regions did not receive any water.

Below 20 cm depth, the photograph shows speckles of Brilliant Blue rather than continuous flow paths. By peeling 0.5-cm thick layers from the surface of the profile plane it became obvious, that these speckles are the cross sections of a network of tortuous flow paths. The map of Brilliant Blue derived from box sampling, which integrates over a 2.2-cm thick layer perpendicular to the surface, shows larger portions of continuous flow paths (A, B, and C in Fig. 1b). All three spots did not appear to be connected to each other and to the main infiltration funnels in the topsoil. They originated from channels created by living tree roots 10 to 30 mm in diameter.

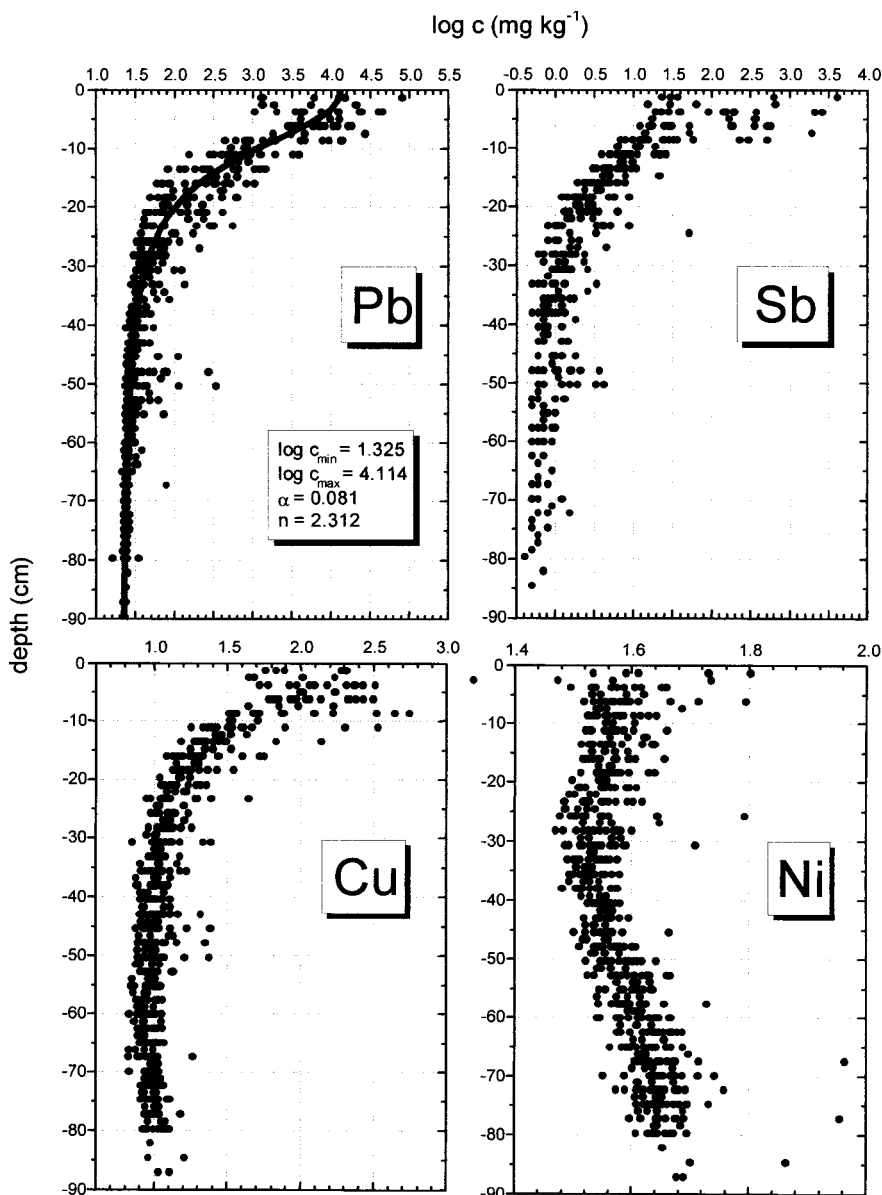
The bromide map (Fig. 1 c), however, shows that all three infiltration funnels combine to one flow path linking spots A, B, and C of the Brilliant Blue map. In line with its more conservative behavior, bromide penetrated through a larger soil volume than Brilliant Blue, delineating the front of the irrigated water. This front is clearly dominated by preferential flow rather than by homogeneous infiltration (Flury et al., 1994).

Two main causes of preferential water flow are generally accepted: (1) fast flow through macropores such as earth-worm burrows, root channels, and other biopores, or through cracks and fissures due to aggregate formation or swelling-shrinking of expandable clay minerals (Beven and Germann, 1982; Booltink and Bouma, 1991; Logsdon, 1995); (2) fingered flow through a homogeneous soil matrix due to wetting front instabilities initiated by differences in water content, wettability of solid surfaces, trapped air, textural boundaries, or by heterogeneous infiltration at the soil

surface, e.g., caused by microtopography or by inhomogeneous distribution of hydrophobic plant litter (Dekker and Ritsema, 1996; Glass et al., 1989; Selker et al., 1992). The observed preferential flow pattern combines both sources, heterogeneous infiltration at the surface due to surface topography, and macropore flow. Brilliant Blue traveled along these preferential flow paths down to 70 cm. Considering the proximity of this deepest spot to the underlying gravelly Cv horizon, there seems to be a substantial risk for the breakthrough of surface contaminants into the groundwater.

### 3.2 Spatial variability of metal contaminants

Fig. 1 d shows that the top 10 cm of the profile are highly contaminated with Pb. The median concentration is  $4.5 \text{ g kg}^{-1}$ , but concentrations as high as  $80.9 \text{ g kg}^{-1}$  were measured (Tab. 2). The Pb concentration rapidly declines with depth, approaching background values of  $20 - 25 \text{ mg kg}^{-1}$  (Fig. 2),



**Figure 2:** Depth profiles of log-transformed concentrations of Pb, Sb, Cu, and Ni displaying all values of the grid sampling ( $n = 626$ ). The depth function of  $\log(\text{Pb})$  is fitted with Eq. 1. Fitted coefficients are given in the inset.

**Abbildung 2:** Tiefenprofile der logarithmieren-ten Pb-, Sb-, Cu- und Ni-Konzentrationen aller Rastermesspunkte ( $n = 626$ ). Die Tiefenfunktion von  $\log(\text{Pb})$  ist mit Eq. 1 gefittet.

that correspond to background values of acidic, silicate-rich bedrocks (Davies, 1995). Of the other metals deposited by the bullets, Sb reaches concentrations as high as  $4.0 \text{ g kg}^{-1}$ , and Cu reaches  $0.55 \text{ g kg}^{-1}$ . However, within 40 cm depth concentrations drop to background levels of  $0.4 \text{ mg Sb kg}^{-1}$  and  $9.4 \text{ mg Cu kg}^{-1}$  (Fig. 2) (Baker and Senft, 1995; Edwards et al., 1995). Although bullets contain also a small amount of Ni ( $5 \text{ g kg}^{-1}$ ), Ni is hardly enriched in the topsoil, with median concentrations around  $40 \text{ mg kg}^{-1}$  in both top- and subsoil (Tab. 2, Fig. 2).

In contrast to the tracer maps, the two-dimensional distribution of Pb (Fig. 1) is dominated by a small variability in horizontal direction and large variability in vertical direction. However, below the most heavily contaminated layer, several spots with elevated Pb concentrations (B, C) coincide with preferential flow patterns. Hence, Pb distribution is mainly explained by soil depth, but additional variability is caused by preferential flow.

In spite of the high Pb concentrations in the topsoil, only two bullets were found in the upper 5 cm of the profile. Since the location of the site just behind the stop butt ensures that deposition of bullets is the predominant cause of the contamination, bullets seem to corrode and dissolve rapidly. This may be due to the low pH, which promotes corrosion and dissolution in general, and due to the high amount of soil organic matter, which may form strong sorption complexes with  $\text{Pb}^{2+}$  and hence create a strong sink for  $\text{Pb}^{2+}$  (Semlali et al., 2001; Wang and Benoit, 1996; Xia et al., 1997). Correspondingly, the strong sorption inhibits the migration of  $\text{Pb}^{2+}$  down the profile, explaining the steep gradient. However, despite the generally strong retention in the topsoil, the elevated Pb concentrations at the spots B and C indicate that exceptional far-reaching transport of Pb into the subsoil occurs in spatially confined areas.

**Table 2:** Median, minimum (Min), and maximum (Max) of metal concentrations at two soil depths.

**Tabelle 2:** Median, Minimum (Min) und Maximum (Max) der Metallkonzentrationen in zwei Bodentiefen.

| Depth       |        | Pb                  | Sb     | Cu    | Ni    | Zn    |
|-------------|--------|---------------------|--------|-------|-------|-------|
|             |        | mg kg <sup>-1</sup> |        |       |       |       |
| 0 to 10 cm  | Median | 4462                | 32.2   | 93.5  | 37.5  | 79.8  |
|             | Max    | 80935               | 4022.4 | 552.3 | 114.7 | 128.3 |
|             | Min    | 429                 | 6.2    | 32.0  | 21.3  | 60.3  |
| 60 to 70 cm | Median | 23                  | 0.4    | 9.4   | 43.3  | 71.8  |
|             | Max    | 78                  | 1.2    | 18.2  | 91.8  | 77.3  |
|             | Min    | 21                  | 0.3    | 6.6   | 34.7  | 65.3  |

Concentrations of Sb and Cu were highly correlated with those of Pb ( $\log[\text{Sb}] = -1.55 + 0.908 \log[\text{Pb}]$ ,  $r=0.95$ ,  $p<0.001$ ; and  $\log[\text{Cu}] = 0.33 + 0.444 \log[\text{Pb}]$ ,  $r=0.95$ ,  $p<0.001$ ). The ratio of Sb to Pb corresponds to the original mass ratio in the bullets ( $\text{Sb/Pb} = 0.02$ ). Maps of Sb and Cu were similar to that of Pb, showing a steep gradient along depth (not shown). Both, Sb and Cu were elevated at spot B, but only Cu was elevated at spot C. These findings indicate that the retention

behavior of these two metals is very similar to that of Pb for the investigated soil with low pH and high organic matter content. In fact, a strong sorption of Cu by soil organic matter similar to that of Pb is well known (Xia et al., 1997). While weathering reactions of Sb in soil have been poorly investigated, there is also some indication that sorption to soil organic matter may play a significant role for the retention of Sb (Pilarski et al., 1995; Rognerud and Fjeld, 2001).

### 3.3 Preferential flow and Pb distribution

Distribution of Pb was dominated by a steep gradient with depth, which makes a statistical assessment of the additional influence of preferential flow difficult. In order to eliminate this strong depth influence, we fitted the depth distribution of the total Pb concentration with a symmetrical, sigmoidal line shape, which links constant topsoil values ( $c_{\max}$ ) with constant background values ( $c_{\min}$ , Fig. 2):

$$\log c = \log c_{\min} + (\log c_{\max} - \log c_{\min}) \left( \frac{1}{1 + (\alpha d)^n} \right) \quad (1)$$

where  $c$  is the fitted Pb total concentration,  $d$  is soil depth in cm, and  $\alpha$  and  $n$  are free fitting parameters.

This function was used to simulate the typical shape of a reactive tracer advancing through a homogeneous porous medium by convection-dispersion. Deviations from the sigmoidal shape, especially a leading tail of the concentration distribution, indicate the existence of rapid travel paths with an impeded mass transfer to and from the surrounding matrix. In fact, spots B and C can be easily recognized in the one-dimensional depth distribution curves (Fig. 2). To test whether such deviations are statistically significant, we investigated the correlation between the deviation of Pb concentrations from the fit (Pb residuals) and tracer concentrations (Tab. 3). The correlation coefficients between Pb residuals and the greenness parameter  $-a^*$  showed a strong depth dependency: while there was no correlation in the uppermost 25 cm, correlation steadily increased with depth. That is, the influence of preferential water flow on the spatial distribution of Pb becomes more important with soil depth. The correlations with Br are generally smaller, suggesting that a reactive tracer like Brilliant Blue is better suited to mark the preferential flow paths relevant for Pb migration.

Thus, in spite of the highly heterogeneous water flow in the whole soil profile, only the preferential flow paths below 20 cm play a significant role in explaining the spatial distribution of Pb. The preferential flow paths below 20 cm are associated with roots surrounded by relatively wide root channels, which represent large macropores for water flow. Since only stained root channels had elevated Pb concentrations, it can be excluded that a biogenic enrichment by the roots themselves took place. Therefore, macropore flow is indeed the process responsible for Pb transport. Lead may be transported in two forms, as aqueous  $\text{Pb}^{2+}$  or bound by colloidal particles (Kretzschmar et al., 1999). Due to the low pH, sorption to Mn or Fe oxides seems to be unlikely, but humic colloids may still bind  $\text{Pb}^{2+}$  by complex formation (Davies et al., 1997;

McKenzie, 1980). However, no decision can be made based on our results, whether Pb has been transported by such mobile humic colloids or as aqueous ion.

Development of a strong spatial correlation between preferential water flow and Pb requires that preferential flow paths are time invariant and last for periods of time long enough to achieve Pb concentrations discernibly higher than in surrounding soil matrix. Since the preferential flow paths follow in many cases the larger tree roots, one may assume they existed for several decades. This is in agreement with results by Bundt et al. (2000) suggesting that preferential flow paths may persist for 20 to 40 years in acidic brown forest soils. In contrast, the topsoil infiltration paths caused by small depressions at the soil surface may change their location frequently, thus preventing the development of strong correlations between Pb and preferential water flow. In fact, the local depressions were part of a deer path as indicated by hoof tracks. This implies that surface topography and infiltration pattern may change with short-term events like a deer passing by.

The Pb concentration declines between the uppermost 20 cm and the subsoil by about three orders of magnitude (Fig. 1d), while the minor decrease of the effective cation exchange capacity suggests a relatively small decline of the retention capacity with depth (Tab. 1). Therefore, the observed concentration gradients suggest a strong retention of Pb and likewise of Cu and Sb, in the topsoil. However, the comparatively small metal concentrations along preferential flow paths do not necessarily mean that metal transport is small and does not present a risk for groundwater pollution. First, the retention characteristics of long lasting preferential flow paths may be substantially weaker than that of the surrounding matrix (Vanderborght et al., 2002). Second, fast transport of dissolved metals or metals bound by colloids, e.g. during the heavy rainfalls typical for Ticino, may cause kinetic limitation of retention processes in the preferential flow paths. Therefore, further studies, including analysis of pore water and investigation of sorption behavior in these flow paths, are necessary to elucidate extent and mechanism of metal transport.

**Table 3:** Correlation coefficients between Brilliant Blue (-a\*) and Br tracer concentrations and the Pb residuals after subtracting the Pb depth function (see Eq. 1 and Fig. 2). A high correlation coefficient indicates a high influence of preferential water flow on Pb distribution for the corresponding soil depth.

**Tabelle 3:** Korrelationskoeffizienten zwischen der Brilliant-Blue-(-a\*) und der Br-Tracerkonzentration und den Pb-Residualkonzentrationen nach Abzug der Pb-Tiefenfunktion (siehe Gl. 1 und Abb. 2). Hohe Korrelationskoeffizienten bedeuten, dass präferenzielle Fließwege einen hohen Einfluss auf die Pb-Verteilung in der entsprechenden Bodentiefe haben.

| Depth       | Correlation coefficients       |                                   |
|-------------|--------------------------------|-----------------------------------|
|             | -a* vs. Pb <sub>residual</sub> | log Br vs. Pb <sub>residual</sub> |
| -5 to 20 cm | -0.01 <sup>(n.s.)</sup>        | -0.04 <sup>(n.s.)</sup>           |
| 21 to 40 cm | 0.29**                         | 0.08 <sup>(n.s.)</sup>            |
| 41 to 60 cm | 0.69**                         | 0.56**                            |
| 61 to 85 cm | 0.82**                         | 0.39**                            |

|             |        |       |
|-------------|--------|-------|
| All samples | 0.12** | 0.10* |
|-------------|--------|-------|

n.s.: not significant

\* significant at  $P < 0.05$

\*\* significant at  $P < 0.01$

## 4 Conclusions

Water flow in the profile was highly heterogeneous, with fingering near the surface induced by heterogeneous infiltration due to surface microtopography, and macropore flow along root channels below 20 cm depth. More than 95 % of total Pb was present in the top 10 cm of the profile, indicating strong retention. While Pb distribution was laterally homogenous in the topsoil, several spots with elevated concentrations ( $> 50 \text{ mg kg}^{-1}$ ) were observed below 40 cm depth. These spots coincided with elevated tracer concentrations, suggesting that preferential flow was responsible for the transport of anthropogenic Pb to the subsoil. Antimony and Cu, which were released together with Pb by the corrosion of bullets, showed a similar distribution pattern, suggesting similar retention and transport mechanisms for all three metals.

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## References

- Adriano, D. C. (1986): Trace Elements in the Terrestrial Environment. Springer Verlag, New York, 356 p.
- Anonymous (1997): Wegleitung: Bodenschutz- und Entsorgungsmassnahmen bei 300 m Schiessanlagen. Generalsekretariat EMD, Bundesamt für Umwelt, Wald und Landschaft (BUWAL), Bern, 128 p.
- Astrup, T., Boddum, J. K., and Christensen, T. H. (1999): Lead distribution and mobility in a soil embankment used as a bullet stop at a shooting range. Journal of Soil Contamination 8, 653–665.
- Baker, D. E., and Senft, J. P. (1995): Copper. In B. J. Alloway: Heavy Metals in Soils. Blackie Academic & Professional, London, pp. 179–205.
- Balikungeri, A., and Haerdi, W. (1988): Complexing abilities of hydrous manganese oxide surfaces and their role in the speciation of heavy metals. Intern. J. Environ. Anal. Chem. 34, 215–225.
- Basunia, S., and Landsberger, S. (2001): Contents and leachability of heavy metals (Pb, Cu, Sb, Zn, As) in soil at the Pantex firing range, Amarillo, Texas. J Air Waste Manage 51, 1428–1435.
- Beven, K., and Germann, P. (1982): Macropores and water flow in soils. Water Resour. Res. 18, 1311–1325.
- Blaser, P., Kernebeek, P., Tebbens, L., Van Breemen, N., and Luster, J. (1997): Cryptopodzolic soils in Switzerland. Eur. J. Soil Sci. 48, 411–423.
- Booltink, H. W. G., and Bouma, J. (1991): Physical and morphological characterization of bypass flow in a well-structured clay soil. Soil Sci. Soc. Am. J. 55, 1249–1254.

- Braun, U., Pusterla, N., and Ossent, P. (1997): Lead poisoning of calves pastured in the target area of a military shooting range. *Schweizer Archiv für Tierheilkunde* 139, 403–407.
- Bruell, R., Nikolaidis, N. P., and Long, R. P. (1999): Evaluation of remedial alternatives of lead from shooting range soil. *Environ. Eng. Sci.* 16, 403–414.
- Bundt, M., Albrecht, A., Froidevaux, P., Blaser, P., and Flühler, H. (2000): Impact of preferential flow on radionuclide distribution in soil. *Environ. Sci. Technol.* 34, 3895–3899.
- Bundt, M., Jaggi, M., Blaser, P., Siegwolf, R., and Hagedorn, F. (2001a): Carbon and nitrogen dynamics in preferential flow paths and matrix of a forest soil. *Soil Sci. Soc. Am. J.* 65, 1529–1538.
- Bundt, M., Widmer, F., Pesaro, M., Zeyer, J., and Blaser, P. (2001b): Preferential flow paths: Biological 'hot spots' in soils. *Soil Biol. Biochem.* 33, 729–738.
- Chen, M., Daroub, S. H., Ma, L. Q., Harris, W. G., and Cao, X. D. (2002): Characterization of lead in soils of a rifle/pistol shooting range in central Florida, USA. *Soil Sediment Contam.* 11, 1–17.
- Cotter-Howells, J. D., Champness, P. E., Charnock, J. M., and Patrick, R. A. D. (1994): Identification of pyromorphite in mine-waste contaminated soils by ATEM and EXAFS. *Eur. J. Soil Sci.* 45, 393–402.
- Craig, J. R., Rimstidt, J. D., Bonnaillon, C. A., Collins, T. K., and Scanlon, P. F. (1999): Surface water transport of lead at a shooting range. *B. Environ. Contam. Tox.* 63, 312–319.
- Davies, B. E. (1995): Lead. In B. J. Alloway: *Heavy Metals in Soils*. Blackie Academic & Professional, London, pp. 206–223.
- Davies, G., Fataftah, A., Cherkasskiy, A., Ghabbour, E. A., Radwan, A., Jansen, S. A., Kolla, S., Paciolla, M. D., Sein, L. T., Buermann, W., Balasubramanian, M., Budnick, J., and Xing, B. (1997): Tight metal binding by humic acids and its role in biomineralization. *J. Chem. Soc., Dalton Trans.*, 4047–4060.
- Dekker, L. W., and Ritsema, C. J. (1996): Preferential flow paths in a water repellent clay soil with grass cover. *Water Resour. Res.* 32, 1239–1249.
- Desaules, A., and Dahinden, R. (1995): Untersuchung von Blei- und Kupfergehalten in Bodenprofilen im Zielgelände von Schiessanlagen, Eidg. Forschungsanstalt für Agrikulturchemie und Umwelthygiene (FAC), Liebefeld-Bern, 145 p.
- Edwards, R., Lepp, N. W., and Jones, K. C. (1995): Other less abundant elements of potential environmental significance: Sb, Au, Mo, Ag, Tl, Sn, U, V. In B. J. Alloway: *Heavy Metals in Soils*. Blackie Academic & Professional, London, pp. 306–352.
- Ellenberg, H. (1986): *Vegetation Mitteleuropas mit den Alpen in ökologischer Sicht*. 4<sup>th</sup> edn., Ulmer, Stuttgart, 989 p.
- Fahrenhorst, C., and Renger, M. (1990): The effects of very large amounts of lead, antimony and arsenic in agricultural soils in the vicinity of shooting ranges. *VDLUFA Kongressberichte* 32, 827–830.
- Flury, M., and Flühler, H. (1995): Tracer characteristics of Brilliant Blue FCF. *Soil Sci. Soc. Am. J.* 59, 22–27.
- Flury, M., Flühler, H., Jury, W. A., and Leuenberger, J. (1994): Susceptibility of soils to preferential flow of water – A field-study. *Water Resour. Res.* 30, 1945–1954.
- Flury, M., Leuenberger, J., Studer, B., and Flühler, H. (1995): Transport of anions and herbicides in a loamy and a sandy field soil. *Water Resour. Res.* 31, 823–835.
- Forrer, I., Papritz, A., Kasteel, R., Flühler, H., and Luca, D. (2000): Quantifying dye tracers in soil profiles by image processing. *Eur. J. Soil Sci.* 51, 313–322.
- Glass, R. J., Steenhuis, T. S., and Parlange, J. Y. (1989): Mechanism for finger persistence in homogeneous, unsaturated, porous media: Theory and verification. *Soil Sci.* 148, 60–70.
- Jørgensen, S. S., and Willems, M. (1987): The transformation of lead pellets in shooting-range soils. *Ambio* 16, 11–15.
- Kabata-Pendias, A., and Pendias, H. (1992): *Trace Elements in Soils and Plants*. 2<sup>nd</sup> edn., CRC Press, 365 p.
- Kretzschmar, R., Borkovec, M., Grolimund, D., and Elimelech, M. (1999): Mobile subsurface colloids and their role in contaminant transport. *Adv. Agron.* 66, 121–194.
- Lewis, L. A., Poppenga, R. J., Davidson, W. R., Fischer, J. R., and Morgan, K. A. (2001): Lead toxicosis and trace element levels in wild birds and mammals at a firearms training facility. *Arch. Environ. Contam. Toxicol.* 41, 208–214.
- Lin, Z. (1996): Secondary mineral phases of metallic lead in soils of shooting ranges from Örebro County, Sweden. *Environmental Geology* 27, 370–375.
- Lin, Z., Comet, B., Qvarfort, U., and Herbert, R. (1995): The chemical and mineralogical behaviour of Pb in shooting range soils from central Sweden. *Environmental Pollution* 89, 303–309.
- Lindsay, W. L. (1979): *Chemical equilibria in soils*. John Wiley & Sons, New York, 449 p.
- Logsdon, S. D. (1995): Flow mechanisms through continuous and buried macropores. *Soil Sci.* 160, 237–242.
- Manceau, A., Charlet, L., Boisset, M. C., Didier, B., and Spadini, L. (1992): Sorption and speciation of heavy metals on hydrous Fe and Mn oxides. From microscopic to macroscopic. *Appl. Clay Sci.* 7, 201–223.
- Manceau, A., Boisset, M. C., Sarret, G., Hazemann, J. L., Mench, M., Cambier, P., and Prost, R. (1996): Direct determination of lead speciation in contaminated soils by EXAFS spectroscopy. *Environ. Sci. Technol.* 30, 1540–1552.
- Manninen, S., and Tanskanen, N. (1993): Transfer of lead from shotgun pellets to humus and three plant species in a Finnish shooting range. *Arch. Environ. Contam. Toxicol.* 24, 410–414.
- McKenzie, R. M. (1980): The adsorption of lead and other heavy metals on oxides of manganese and iron. *Aust. J. Soil Res.* 18, 61–73.
- Miyazaki, T. (1993): *Water Flow in Soils*, Marcel Dekker, New York, 296 p.
- Murray, K., Bazzi, A., Carter, C., Ehler, A., Harris, A., Kopec, M., Richardson, J., and Sokol, H. (1997): Distribution and mobility of lead in soils at an outdoor shooting range. *Journal of Soil Contamination* 6, 79–93.
- Pilarski, J., Waller, P., and Pickering, W. (1995): Sorption of Antimony species by humic-acid. *Water Air Soil Poll.* 84, 51–59.
- Rognerud, S., and Fjeld, E. (2001): Trace element contamination of Norwegian lake sediments. *Ambio* 30, 11–19.
- Rooney, C. P., and McLaren, R. G. (1999): Leaching of lead from soils contaminated with lead shot. In "Proc. 5th Intern. Conf. On The Biogeochem. Of Trace Elements", Vienna, pp. 776–777.
- Sarret, G., Manceau, A., Cuny, D., Van Haluwyn, C., Deruelle, S., Hazemann, J.-L., Soldo, Y., Eybert-Berard, L., and Menthonnex, J.-J. (1998): Mechanisms of lichen resistance to metallic pollution. *Environ. Sci. Technol.* 32, 3325–3330.
- Scheinost, A. C., and Schwertmann, U. (1999): Color identification of iron oxides and hydroxysulfates – Use and limitations. *Soil Sci. Soc. Am. J.* 63, 1463–1471.
- Scheinost, A. C., Abend, S., Pandya, K. I., and Sparks, D. L. (2001): Kinetic controls on Cu and Pb sorption by ferrihydrite. *Environ. Sci. Technol.* 35, 1090–1096.

- Selker, J. S., Steenhuis, T. S., and Parlange, J. Y. (1992): Wetting front instability in homogeneous sandy soils under continuous infiltration. *Soil Sci. Soc. Am. J.* 56, 1346–1350.
- Semlali, R. M., Van Oort, F., Deniaix, L., and Loubet, M. (2001): Estimating distribution of endogeneous and exogeneous Pb in soils by using Pb isotopic ratios. *Environ. Sci. Technol.* 35, 4180–4188.
- Turpeinen, R., Salminen, J., and Kairesalo, T. (2000): Mobility and bioavailability of lead in contaminated boreal forest soil. *Environ. Sci. Technol.* 34, 5152–5156.
- Vanderborght, J., Gähwiler, P., and Fühler, H. (2002): Identification of transport processes in soil cores using fluorescent tracers. *Soil Sci. Soc. Am. J.* in press.
- Wang, E. X., and Benoit, G. (1996): Mechanisms controlling the mobility of lead in the Spodosols of a Northern hardwood forest ecosystem. *Environ. Sci. Technol.* 30, 2211–2219.
- Wixson, B. G., and Davies, B. E. (1994): Guidelines for lead in soil proposal of the Society for Environmental Geochemistry and Health. *Environ. Sci. Technol.* 28, A26–A31.
- Xia, K., Bleam, W., and Helmke, P. A. (1997): Studies of the nature of  $\text{Cu}^{2+}$  and  $\text{Pb}^{2+}$  binding sites in soil humic substances using X-ray absorption spectroscopy. *Geochim. Cosmochim. Acta* 61, 2211–2221.